

Matthew T. Frederick, Ph.D.

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RELATED EXPERIENCE

Editor for American Journal Experts

Mar. 2013 – Present

- Self-directed and able to meet deadlines on short notice.
- Rapidly review, interpret, and prepare technical manuscripts covering virtually all subjects within the physical sciences and select engineering and business disciplines.
- Strong writing skills and ability to understand intended meaning despite ambiguity.

Postdoctoral Research, Northwestern University

Dec. 2012 – Apr. 2013

- Planned the long-term strategies for the students continuing my graduate research.
- Anticipated possible issues and provided potential resolutions for these research plans.

Graduate Research, Northwestern University

Sept. 2008 – Nov. 2012

- Conceived, designed, performed, and interpreted the major component of my advisor's tenure packet, which has resulted in many high-impact publications (*h-index*: 10; *total citations*: 275).
- Excellent ability to work in a team and analyze, compile, and integrate data from scientists with varied backgrounds due to experience collaborating with a broad range of scientists and engineers on research projects.
- Founding member of a research group, which required independence, perseverance, and resourcefulness to establish lab protocols and build a knowledge base with minimal guidance and limited resources.
- Provided senior leadership, management, and education for all new classes of incoming graduate and undergraduate students during Ph.D. career.
- Hierarchical Materials Cluster Program Fellow – collaborated on multilayered energy production research and integrated technical information from multiple scientific and engineering disciplines.

Kellogg School of Management, Northwestern University

Summer 2012

Management for Scientists and Engineers

Evanston, IL

- Course designed to educate exceptional Ph.D. students about typical business and management concepts covered during an MBA.
- Course topics included environmental externalities, risk management, costs associated with worker safety, public/private partnerships, leadership/team management, conflict resolution, contract negotiation, and governmental policies and regulations.

Chicago Council on Science and Technology

Dec. 2011 – Apr. 2013

- Presented written summaries of scientific topics to encourage the Chicago community to engage with scientific issues (including environmental and energy policy).

Northwestern University Politics and Policy Committee Member

Aug. 2012 – Apr. 2013

- Disseminated important current events to the Northwestern University Community and elucidated their consequences on the global political stage.

Science in the Classroom

Sept. 2010 – Jun. 2010

- Fostered excitement about science in young children through entertaining and educational demonstrations.

EDUCATION

Northwestern University

Ph.D. Physical/Analytical Chemistry

- Thesis: *Mechanistic Description of Surfactant-Induced Exciton Delocalization in Semiconductor Quantum Dots*
- Advisor: Professor Emily Weiss
- GPA: 3.6; GRE: Math: 800/800 Verbal: 580/800 Writing: 6/6

Nov. 2012

Evanston, IL

Kellogg School of Management, Northwestern University

Management for Scientists and Engineers

Summer 2012

Evanston, IL

University of North Carolina at Chapel Hill

B.S. Chemistry

- Advisor: Professor Royce Murray
- GPA: 3.7; SAT: Math: 730/800 Verbal: 750/800

May 2008

Chapel Hill, NC

TECHNICAL SKILLS

Scientific: surfaces, nanostructures, semiconductors, colloids, electron transfer/transport, photophysics, material science, electronic structure of heterogeneous interfaces, electrochemistry

Communication: writing, editing, oral presentation, one-on-one and small group instruction, delegation, mentoring, adaptability, collaboration, flexibility

Laboratory: XPS, TEM, SEM, AFM, NMR, DOSY, Electrochemistry, Photoconductivity, MALDI, HPLC-MS, ICP-AES, FTIR, Fluorescence, UV/Visible Absorption

Computer: Full Microsoft Office Suite, OriginPro, QChem, Mathematica, LabVIEW, and Limited Python, Matlab, and Java experience

PRESENTATIONS

The Edge of Confinement: An Organic Ligand to Determine the Radius of Individual Charge Carriers

Clusters, Nanocrystals, & Nanostructures Gordon Research Conference

July 2011

South Hadley, MA

Relaxation of Exciton Confinement through Partial Hole Transfer to a Conjugated Dithiocarbamate Ligand

NSF-MRSEC Workshop on Transport in Nanoengineered Materials

Sept. 2010

Chicago, IL

Relaxation of Exciton Confinement through Partial Hole Transfer to a Conjugated Dithiocarbamate Ligand

Electron Donor-Acceptor Gordon Research Conference

Aug. 2010

Newport, RI

INTERESTS

Biking, hiking, skiing, sailing, traveling, philosophy, home brewing, surfing